

Shuo Zhang (Clyde)

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Undergraduate Researcher (Sophomore) · Prospective PhD Candidate
Harbin Institute of Technology (HIT)

★ GPA 3.8 / 4.0

◆ IELTS 7.5

Research Interests

- **Ultra-precision displacement and angle measurement** — autocollimator & interferometer-based metrology
- **Adaptive optics** — wavefront sensing and correction for imaging/laser systems
- **Metasurfaces & micro/nano optics** — subwavelength flat-optics beyond refractive limits
- **Nonlinear optics** — Kerr nonlinearities, temporal solitons & quasi-phase-matching
- **Computational imaging** — super-resolution microscopy & bioimaging, recovering lost information

Education

Heihe No.1 High School

Heilongjiang, China · 2021 – 2024

B.Eng. in Optical Engineering (expected 2028)

Harbin Institute of Technology, School of Future Technology · 2024 – Present · GPA 3.8/4.0

- People's Scholarship, awarded multiple times

Research Experience

Ultra-Precision Displacement & Angle Measurement

2024 – Present

Developed autocollimator- and interferometer-based techniques for nanometer-scale displacement and microradian-level angle metrology. Modeled systematic error sources in optical measurement systems and proposed compensation methods. Published first-author work on lens systematic errors in autocollimators (*Sensors*, MDPI).

Stray-Light Suppression for Optical Systems

2025 – Present

Proposed a universal stray-light suppression method applicable to all-spherical optical systems. Presented at ICSA (International Conference on Systems and Automation).

Nonlinear Optics & Metasurface Research

2025 – Present

Investigating Kerr nonlinearities, temporal solitons, and quasi-phase-matching for frequency conversion, plus subwavelength metasurface design for compact beam-shaping and imaging. Five manuscripts in preparation (*Optics Express* & EI-indexed journal).

Computational Imaging & Bioimaging

2025 – Present

Combining optics with reconstruction algorithms to image beyond the diffraction limit — super-resolution microscopy and computational bioimaging, where inverse-problem methods recover information lost along the optical path. Bridges metrology, nonlinear optics, and life-science imaging.

Bachelor's Thesis — Three-Degree-of-Freedom Pose Measurement Based on Heterodyne Interference and Autocollimation

Expected 2028

Capstone project extending the heterodyne-interference and autocollimation principle to single-beam, three-degree-of-freedom (displacement, pitch/yaw, roll) pose metrology — part of the precision-measurement

line that also produced the pending patent and the under-review *Optics and Laser Engineering* paper.

Honors & Awards

- People's Scholarship, HIT (multiple times)
 - Provincial First Prize, National College Student Mathematical Modeling Contest
 - University-level Outstanding Student (■■■■■)
 - Provincial First Prize, National High School Physics Competition
 - Provincial First Prize, National Undergraduate Electronic Design Contest
 - Provincial First Prize, Chinese Physics Competition for College Students
 - University-level Outstanding League Member (■■■■■)

Publications

Influence of Lens Systematic Errors on Autocollimator Angle Measurement: Theoretical and Experimental Explanations.

S. Zhang et al., *Sensors* (MDPI), 2025. DOI: 10.3390/s25247654

A Universal Stray Light Suppression Method for All-Spherical Optical Systems.

S. Zhang et al., *ICSA (International Conference on Systems and Automation)*, 2025.

Distributed-order dispersion for endpoint-preserving control of temporal-soliton internal modes.

S. Zhang et al., *Optics Express* (under review), 2026.

Fabrication-aware inverse design and performance bounds of binary quasi-phase-matching gratings for frequency-multiplexed sum-frequency generation.

S. Zhang et al., *Optics Express* (under review), 2026.

Kerr-induced renormalization of the Talbot recurrence length in periodic optical fields.

S. Zhang et al., *Optics Express* (under review), 2026.

Stray lights in autocollimator: categories, analysis, and suppression.

S. Zhang et al., *Optics Express* (under review), 2026.

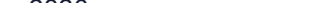
Suppressing sensitivity of assembly deviation of autocollimator by collimator objective design.

S. Zhang et al., *Annual Conference on Instrumentation* (EI-indexed, under review), 2026.

Research on Three-Degree-of-Freedom Pose Measurement Technology Based on Heterodyne Interference and Autocollimation.

S. Zhang et al., *Optics and Laser Engineering* (under review), 2026.

Patents

-  (A Single-Beam Three-Degree-of-Freedom Measurement Method and Device Based on Heterodyne Interferometry and Autocollimation). Pending examination, 2026.

Skills

Python [MATLAB [Zemax/OpticStudio [Optical [Metrology [Adaptive [Optics [Metasurface [Design [Nonlinear [Optics [Computational [Imaging [Bioimaging [Super-resolution [LaTeX [Data [Analysis [Electronic [Prototyping]

Updated July 2026 · Also available at the author's academic website.